

3. Ebony Body (आबनूसी देह)

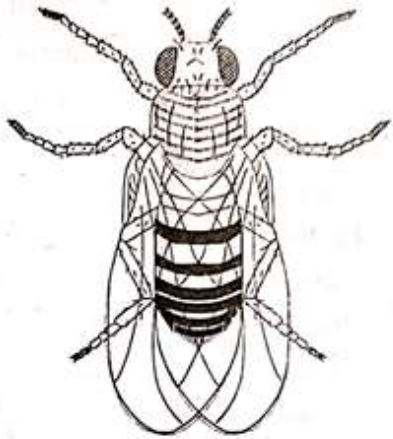
- (i) The body color of the adult is black.
- (ii) The eyes are highly reduced.
- (iii) This mutation is caused by a change in a gene located at position **70.7** on the **third chromosome**.
- (iv) This is a **dominant autosomal mutant**.

4. Vestigial Winged Drosophila (अवशेषी पंख वाली ड्रोसोफिला)

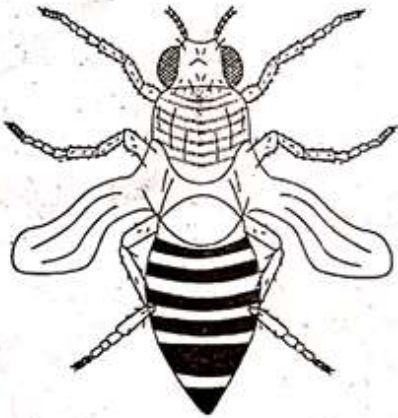
- (i) This is a mutant species of Drosophila.
- (ii) Both wings present on the body are extremely small, underdeveloped, or of the **vestigial** type.
- (iii) This type of mutation occurs due to a change in the gene located at position **67.0** on its chromosome.
- (iv) This is a **recessive** type trait.

5. White-Eyed Drosophila (श्वेत नेत्रों वाला ड्रोसोफिला)

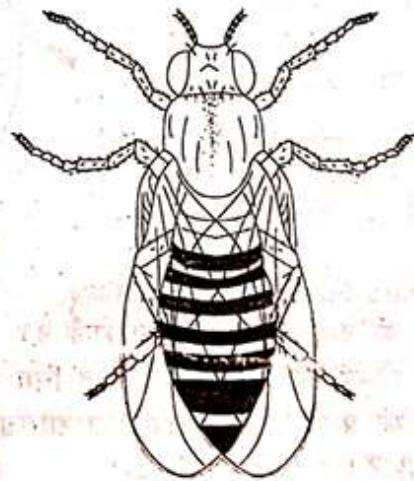
- (i) This is a mutant species of Drosophila.
- (ii) The color of its body is white. (Note: In standard genetics, "white" usually refers to the eye color; the text here notes body color as white/light).
- (iii) Its eye color is **apricot** or **coral white**.
- (iv) This mutation occurs due to a change in the gene located at position **1.5** on the **first chromosome (X-chromosome)**.
- (v) This is a **recessive** type trait.



चित्र 3 : आबनूसी ड्रोसोफिला



चित्र 4 : अवशेषी पंखों वाला ड्रोसोफिला



चित्र 5 : श्वेत नेत्र वाला ड्रोसोफिला

Objective: To study the salivary gland chromosomes of *Drosophila*

(उद्देश्य: ड्रोसोफिला के लार ग्रंथि गुणसूत्रों का अध्ययन करना)

Salivary Gland Chromosomes or Polytene Chromosomes

- These chromosomes were first seen in the cells of the salivary glands of **Diptera** insects; therefore, they are called **salivary gland chromosomes**.
- This slide is of a salivary gland chromosome; they are also called **polytene chromosomes**.
- They were first discovered by **E.G. Balbiani** in **1881**.
- These chromosomes are formed by the representation of a pool of many chromatids, whereas in normal chromosomes, their number is two.
- These chromosomes are a **thousand times larger** than single-pool chromosomes and are visible during the **interphase**.
- Special types of **transverse bands** are found on these chromosomes. These are dark and light in color. These bands are found in an alternating pattern.
- These bands are genetically active and DNA is found in abundance in them.
- At short distances on these chromosomes, several swollen structures are found, which are called **Balbani rings** (or puffs).
- These rings are formed by lateral swelling in the band region.
- The formation of these rings on these chromosomes is under the control of **specific genes**.

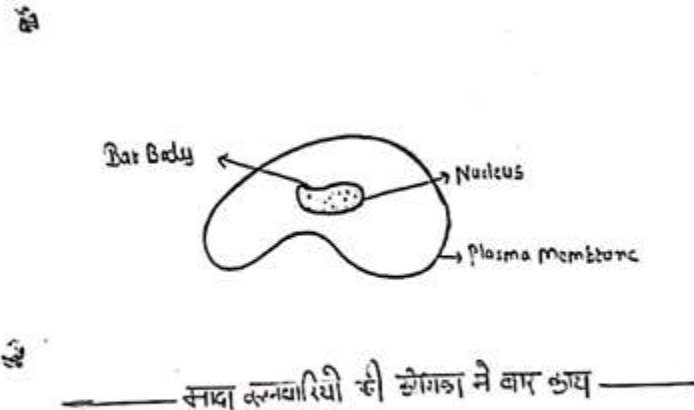
This page moves from fruit flies to human genetics, specifically the study of **Barr bodies**. Here is the exact translation of your notes:

Objective: To study Barr bodies in human buccal epithelial cells.

(उद्देश्य: मानव मुख उपकला कोशिकाओं में बार काय का अध्ययन)

- It was discovered by **Barr and Bertram** in the cells of mammals.
- This body is found in the form of a mass located near the nuclear membrane in the cells of normal female mammals.
- **Barr bodies** are generally found in the cells of female organisms.
- Therefore, they can be used for **sex differentiation**.
- Barr bodies are made of **heterochromatin**; therefore, they are deeply stained.
- Barr bodies are a modified form of the **sex chromosome (X)**; therefore, they are also called **sex chromatin**.
- The number of Barr bodies is always **one less** than the total number of **X chromosomes**. Thus, they are absent in males (who have only one X chromosome).

- Based on the presence of Barr bodies, the **sex of an offspring** can be determined during pregnancy (gestation).



This page introduces the concept of blood groups in humans. Here is the exact translation:

Experiment No. 5

Objective: Identification of blood groups in man

(उद्देश्य: मनुष्य में रक्त समूहों का अभिनिर्धारण या पहचान करना)

Introduction (परिचय)

The credit for the discovery of blood groups in humans goes to **Karl Landsteiner (1900)** and his associates. According to them, the blood of different humans can be classified into **four groups** based on the presence or absence of **antigens** present in the red blood cells or **erythrocytes (R.B.C.)**.

For this reason, sometimes when the blood of one person is **transfused** to another person, the recipient may die due to **agglutination** (clumping) in the blood.

Blood Group Determination Table

(रक्त समूह निर्धारण तालिका)

Antigens on Red Blood Cells	Antibodies in Blood Serum	Blood Group
A	Anti-B (β)	"A"
B	Anti-A (α)	"B"
A and B present	No Antibodies	"AB"
Both A and B absent	Anti-A and Anti-B antibodies present	"O"

(Note: There is a typo in your printed text where it says "B" for the last row; in the ABO system, the absence of both antigens identifies Blood Group "O").

Procedure for Determination of Blood Groups

(रुधिर वर्गों के निर्धारण की प्रक्रिया)

1. Clean the middle finger of the person whose blood is to be tested with cotton soaked in spirit or **90% alcohol** and let the fingertip dry.
2. Prick the fingertip with a sterilized needle and collect one drop of blood into a small vial. This vial should already contain **1 ml of 0.85% Sodium Chloride** solution; shake to mix.
3. Take a slide with **two cavities**. If a single-cavity slide is available, take two slides. Label these cavities **A** and **B** with a glass pencil.
4. Using a fine pipette, take the blood from the vial and place one drop in each of the two cavities (A and B) of the slide.
5. Add one drop of **Anti-A serum** to cavity **A** and one drop of **Anti-B serum** to cavity **B**.
6. Mix them using separate needles or glass rods. After **ten minutes**, examine the blood to see if the process of **clumping or agglutination** has occurred. Record the results as follows

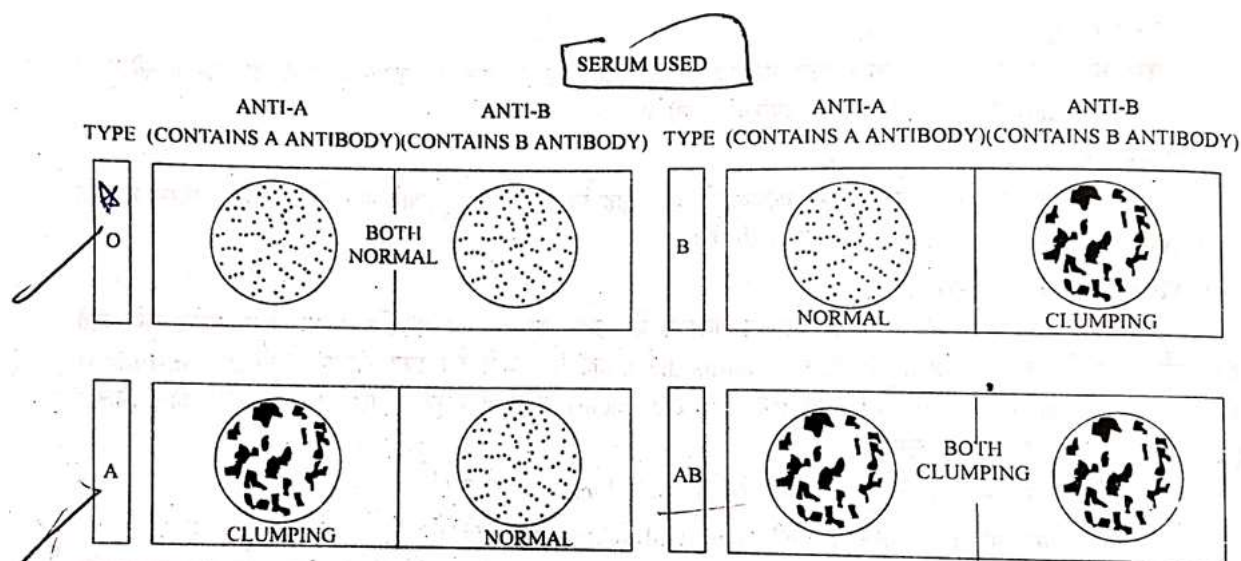


Figure 13: Testing of Blood Groups in Man

(चित्र 13: मनुष्य में रूधिर वर्गों का परीक्षण)

- (i) No clumping in both cavities — Blood group is **O**.
- (ii) Clumping occurred in cavity A — Blood group is **A**.
- (iii) Clumping occurred in cavity B — Blood group is **B**.
- (iv) Clumping occurred in both cavities A and B — Blood group is **AB**.

Experiment - 10

Determination of pH of Soil Sample

Object: To study the pH of soil.

(उद्देश्य: मृदा के pH का अध्ययन।)

Measurement of soil pH can be done by two methods:

1. **Electronic measurement**
2. **Colorimetric measurement**

(1) Electronic Measurement:

Principle: The negative log of the concentration of hydrogen or hydronium ions in a solution is called pH ($\text{pH} = -\log[\text{H}^+]$).

Alternatively, its measurement can be done with the help of **electrodes**. Among these electrodes, one is sensitive to H^+ ions, while the second electrode is sensitive to OH^- . The potential of both can be seen on a scale.

Since the generated potential is affected by temperature and some other factors, it is controlled for temperature. Additionally, the meter is set (calibrated) using a known pH solution.

Reagents:

1. **Potassium Hydrogen Phthalate Buffer** ($KHC_8H_4O_4$)
2. **Phosphate Buffer**
3. **Borax Buffer**

Procedure:

1. The pH of soil depends on the **water ratio**. Therefore, if you want to determine the pH of soil at a **saturation level**, take a **50 gm soil sample** and slowly add **distilled water** to it.
 2. When water starts glistening on the surface of the soil, stop adding water. Now, stir the soil with a **glass rod**.
 - 3.
 4. Prepare an **extract** from the saturated soil. For this, a soil-to-water ratio of **1:5 or 1:10** can be taken. For example, take **20 gm of soil in 100 ml of water** and shake it for one hour.
-

(2) Colorimetric Measurement:

In this method, the pH of the soil is determined by the **color change** that occurs in various indicators. For this, any one of the following methods can be adopted:

- **(A) Determining pH using pH paper strips:** Finding pH by comparing the color change of the paper to a standard chart.
- **(B) Determining pH using Indicator Solution:** Adding chemical indicators to the soil extract to observe color changes.
- **(C) Determining pH using the Thiocyanate method:** A specific chemical reaction used to estimate acidity/alkalinity.

This page details a chemistry experiment used to measure the minerals in water. Here is the exact translation of your notes:

Experiment – 11

Water Hardness of Water

Object: To determine the hardness of water by the **EDTA Method**.

(उद्देश्य: EDTA Method द्वारा जल की कठोरता ज्ञात करना।)

Principle: The main cations that cause hardness in water are Ca^{2+} and Mg^{2+} . The amount of other hardness-causing cations is negligible compared to these.

At **pH = 10.0**, Ca and Mg form a **wine-red complex** with a compound called **Eriochrome Black-T (EBT)**. When EDTA solution is added, this complex breaks because EDTA has a higher affinity for these two ions. Consequently, a **blue-colored** complex is formed at the end.

Chemical Equations:

- $\text{Ca/Mg} + \text{Eriochrome Black-T} \rightarrow \text{Wine red Complex}$
- $\text{Ca/Mg (EBT Complex)} + \text{EDTA} \rightarrow \text{Ca/Mg-EDTA} + \text{Eriochrome Black-T}$

*(The release of free EBT results in a **blue-black complex/color**)*

Apparatus:

- Pipette (पिपेट)
 - Flask (फ्लास्क)
 - Funnel (कीप)
 - Beaker (बीकर)
-

Chemicals:

1. **0.01 M EDTA Solution** (Ethylene Diamine Tetraacetic Acid)
2. **Buffer Solution** (Used to maintain the pH at 10.0)

Chemicals (continued):

- (iii) Eriochrome Black-T (indicator)
 - (iv) Sodium Sulfide Solution
-

Procedure (कार्यविधि):

1. Take the water **sample** in a **50 ml** volume flask.
 2. Add **1 ml of buffer solution** to it.
 3. Add **1 ml of Sodium Sulfide solution** to it.
 4. Add **100 mg of Eriochrome Black-T indicator** to the solution. Upon mixing this, the solution turns **wine red** (अंगूरी लाल).
 5. Take **EDTA (0.01M)** in the burette and perform the **titration**.
 6. During titration, note the point at which the color of the solution turns **blue**.
-

Result:

The hardness of the water is **370 mg/lit**.

Observation Table

S.No.	Volume of Sample (ml)	Burette Reading (ml) - Initial	Burette Reading (ml) - Final	Total Volume Used (ml)	Mean (माध्य)
1.	50 ml	0	10.0	10.0	
2.	50 ml	0	6.0	6.0	8.6
3.	50 ml	0	10.0		

This page provides the theoretical background and preparation for the experiment you just showed the calculations for. It uses **Mohr's Method** to find the amount of chloride in water. Here is the translation:

Experiment – 8

Determine the amount of Chloride

Object: To determine the amount of chloride by **Mohr's Method**.

(उद्देश्य: Mohr's Method द्वारा क्लोराइड की मात्रा ज्ञात करना।)

Principle: When Silver Nitrate (AgNO_3) reacts with chloride present in water, a white precipitate of Silver Chloride (AgCl) is formed.



When all the chloride in the water has precipitated, the AgNO_3 becomes available in a free state in the solution. Its presence can be made clear using **Potassium Chromate** (K_2CrO_4) as an indicator. This is because this indicator reacts with AgNO_3 to form silver chromate, which is **red** in color. The appearance of this red color is the **end point**.



(Ag_2CrO_4 is Red)

Apparatus:

- Bottle (बोतल)
 - Burette (ब्यूरेट)
 - Conical Flask (कोनिकल फ्लास्क)
 - Reagents (अभिकर्मक)
-

Chemicals:

1. **0.02 N Silver Nitrate** (AgNO_3)
2. **5% Potassium Chromate** (K_2CrO_4)

(Prepared by mixing 5g of K_2CrO_4 in 100 ml of water)

Procedure (कार्यविधि):

1. Take the **sample** in a plastic or **borosilicate bottle (50 ml)**.
 2. Take the sample in a flask (it will appear **yellow**) and add **2 ml of Potassium Chromate indicator** to it.
 3. Take the **Silver Nitrate ($AgNO_3$) solution** in a burette and **titrate** the sample. The sample changes from a **yellow color to a milky (turbid) color**.
 4. When the **red color becomes permanent**, consider that to be the **end point**.
-

Result:

The amount of dissolved Cl^- (**Chloride**) in the water is **142 mg/L**.

- (5) If the sample is colorless, it should be titrated with **0.05 N Sodium Hydroxide ($NaOH$)** filled in the burette.
- (6) As soon as the color of the solution turns **pink**, stop adding the hydroxide and record the volume of **0.05 N $NaOH$** used.

Result:

The amount of dissolved CO_2 in the water is **66 mg/L**.

Observation Table

S.No.	Volume of Sample (ml)	Burette Reading (ml) - Initial	Burette Reading (ml) - Final	Total Volume Used (ml)	Mean (माध्य)
1.	100 ml	0	2.5	2.5	
2.	100 ml	0	3	3	3

S.No.	Volume of Sample (ml)	Burette Reading (ml) - Initial	Burette Reading (ml) - Final	Total Volume Used (ml)	Mean (माध्य)
3.	100 ml	0	3.5	3.5	

Calculation

The formula used to find dissolved CO_2 in mg/L is:

$$\text{Dissolved } \text{CO}_2 \text{ in mg/L} = \frac{A \times \text{Normality of NaOH} \times 1000 \times 44}{\text{ml of Sample}}$$

Average Volume (A):

$$\text{Average} = \frac{2.5 + 3 + 3.5}{3} = \frac{9.0}{3} = 3.0 \text{ ml}$$

Final Calculation:

(Note: There is a complex denominator in your notes, but the standard simplified math is shown below)

$$\rightarrow \frac{3.0 \times 0.05 \times 1000 \times 44}{100}$$

$$\rightarrow \mathbf{66 \text{ mg/L}}$$

Objective: Differentiation or Qualitative Study of Plankton

(उद्देश्य: प्लवकों का विभेदन अथवा गुणात्मक अध्ययन)

Chemical substances present in any water body represent the quality or status of that water. Their measurement gives us information about the merits and demerits of the water. **Plankton** in water are also affected by the chemical and physical factors of the water; therefore, the quantity and type of plankton are indicators of water quality.

- Some plankton are found in highly polluted water because they are **pollution-tolerant**, while some are found in nearly clean water.
- Some plankton are found only in freshwater, while some can survive in saline (salty) water as well.

In this way, the presence of certain species in the water acts like a mirror to many chemical or physical properties. Some plankton give definitive information about pollution. These are called **bio-indicators**. Genera such as **Euglena**, **Chlamydomonas**, **Oscillatoria**, and **Scenedesmus** are indicators of pollution.

From this fact, it is clear that instead of knowing pollution through various chemical tests of water, the status of the water or the level of pollution can also be determined by the presence of bio-indicators. Similarly, the effectiveness of water treatment can also be checked by observing them. Ecological changes can also be estimated by looking at them.

For the above reasons and due to academic interest or curiosity, we should know how to classify plankton.

Experience is required to be able to differentiate between plant and animal plankton. This experience can be gained by looking at more and more plankton slides and by observing samples. View permanent slides of phytoplankton and zooplankton under a microscope.

If **plant plankton** are multicellular, they exhibit **branching**, or the cells are connected in a sequence like beads in a necklace, or they form clusters. This type of structure is not found in higher classes (Metazoa) beyond Protozoa.

After classifying a plankton into either the plant or animal kingdom, it must be classified further. For this task, help can be taken from **taxonomic keys** or **pictorial keys**. The deeper the level of identification required (down to lower taxa), the more knowledge and experience are needed. However, it is generally not difficult to identify an animal down to its class, order, family, and sometimes even genus.

Study the permanent slides made for this task. Pay special attention to the characteristics because of which they are placed in a specific species, genus, or family. These traits are often quantitative as well; therefore, one must also learn the method of measurement using a microscope. Refer to

appropriate taxonomic literature for these tasks. Given below are the identifying characteristics of the major aquatic animal **phyla**.

(i) Protozoa (प्रोटोजोआ)

- Most organisms are **unicellular**, though some are **colonial**.
- Except for a few species (like *Euglena*), chlorophyll is not found in the rest.
- **Cilia, flagella, or pseudopodia** (कूट पाद) may be present for locomotion (movement).